Digital Electronics

Report on
Boolean Algebra and K-Maps

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Year / Group	2nd Year, Group-36
Subject	Digital Electronics
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PPT Link	Click here to open the PPT on GitHub

Full link (copy and paste into a browser if the click does not work):
https://github.com/anushka2006sharma-ui/Boolean-Algebra-and-K-maps/blob/main/Boolean%20Algebra%20and%20K-Maps.pptx

1. Introduction

Boolean algebra is the algebra of two values, 0 and 1 (false and true), introduced by George Boole in 1854. It is the mathematical basis of digital electronics: variables represent signals, and expressions describe the gates that process them. The three basic operations are AND (.), OR (+) and NOT (').

2. Logic Gates

Gate	Expression	Output is 1 when
AND	A.B	both inputs are 1
OR	A+B	at least one input is 1
NOT	A'	input is 0
NAND	(A.B)'	not both inputs are 1
NOR	(A+B)'	both inputs are 0
XOR	A xor B	inputs differ

3. Laws of Boolean Algebra

Law	AND form	OR form
Identity	A.1 = A	A+0 = A
Null	A.0 = 0	A+1 = 1
Idempotent	A.A = A	A+A = A
Complement	A.A' = 0	A+A' = 1
Commutative	A.B = B.A	A+B = B+A
Associative	(A.B).C = A.(B.C)	(A+B)+C = A+(B+C)
Distributive	A.(B+C) = AB+AC	A+BC = (A+B)(A+C)
Absorption	A.(A+B) = A	A+AB = A

4. De Morgan's Theorems

(A.B)' = A' + B' and (A+B)' = A'.B'

Example: F = AB + AB' = A(B+B') = A.1 = **A** (distributive, complement, identity).

5. SOP and POS Forms

Sum of Products (SOP) is an OR of AND terms (minterms), e.g. F = A'B + AB'.

Product of Sums (POS) is an AND of OR terms (maxterms), e.g. F = (A+B)(A'+B').

Both describe the same truth table.

6. Karnaugh Maps (K-maps)

A K-map is a grid form of the truth table that simplifies Boolean expressions visually. Rows and columns follow Gray code (00, 01, 11, 10) so neighbouring cells differ by one bit. A 2, 3 or 4-variable map has 4, 8 or 16 cells.

Rules: group adjacent 1s in powers of 2 (1, 2, 4, 8); make groups as large as possible; groups may wrap around the edges; don't-care cells may be used if they enlarge a group.

Example: F(A,B,C) = Sum of minterms (1, 3, 6, 7)

A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

Group m1, m3 gives A'C. Group m6, m7 gives AB. Simplified result: **F = A'C + AB**.

7. Applications and Uses

- **Circuit design:** fewer gates, lower cost and power.
- **CPU and ALU:** adders, comparators and control logic.
- **Memory:** address decoders and selectors.
- **Multiplexers and demultiplexers:** data routing.
- **Error detection:** parity checkers using XOR.
- **Software and AI:** conditions, search queries and decision logic.

8. Conclusion

Boolean algebra gives the rules for binary logic. Its laws and De Morgan's theorems simplify expressions, and K-maps make that simplification fast and visual, resulting in smaller and cheaper digital circuits.